

```

import cv2
import numpy as np
import matplotlib.pyplot as plt
from google.colab import files

# Upload grayscale image
print("Upload a grayscale image")
uploaded = files.upload()

filename = list(uploaded.keys())[0]

# Read image in grayscale
img = cv2.imread(filename, cv2.IMREAD_GRAYSCALE)

# Function for quantization
def quantize(image, levels):
    step = 256 // levels
    return (image // step) * step

# Different quantization levels
q256 = img
q64 = quantize(img, 64)
q16 = quantize(img, 16)
q4 = quantize(img, 4)

# Display images
plt.figure(figsize=(12,8))

titles = ["Original (256 Levels)",
          "64 Levels",
          "16 Levels",
          "4 Levels"]

images = [q256, q64, q16, q4]

for i in range(4):
    plt.subplot(2,2,i+1)
    plt.imshow(images[i], cmap='gray')
    plt.title(titles[i])
    plt.axis("off")

plt.show()

# Output
print("\n===== QUANTIZATION REPORT =====")
print("Original Image      : 256 Gray Levels")
print("Image 1                : 64 Gray Levels")
print("Image 2                : 16 Gray Levels")
print("Image 3                : 4 Gray Levels")
print("-----")
print("Observation:")
print("- Higher gray levels preserve image detail.")
print("- Lower gray levels reduce detail.")
print("- Very low levels create visible intensity bands.")
print("=====")

```

Upload a grayscale image

 No file chosen

Saving Q7.jpg to Q7.jpg

Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

Original (256 Levels)



64 Levels



16 Levels



4 Levels



===== QUANTIZATION REPORT =====

Original Image : 256 Gray Levels

Image 1 : 64 Gray Levels

Image 2 : 16 Gray Levels

Image 3 : 4 Gray Levels

Observation:

- Higher gray levels preserve image detail.
- Lower gray levels reduce detail.
- Very low levels create visible intensity bands.

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